

Team Iterative Processes (SB5-MAI)

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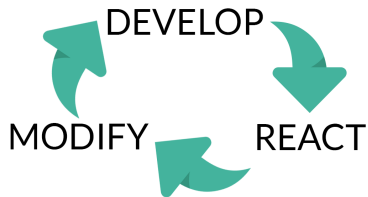
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Team Iterative Processes

- ▶ Most of the software projects require a larger effort than a solo programmer can handle
- ▶ Programmers have to organize themselves into teams

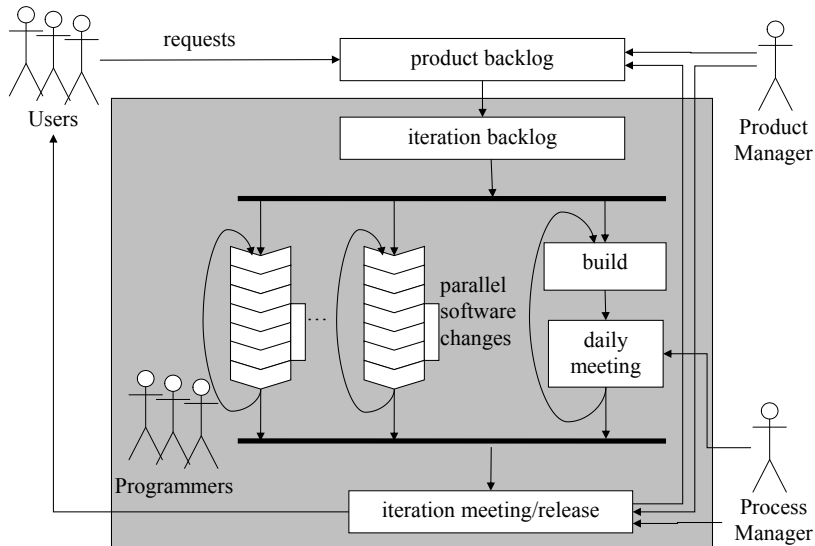


Agile Iterative Processes (AIP)



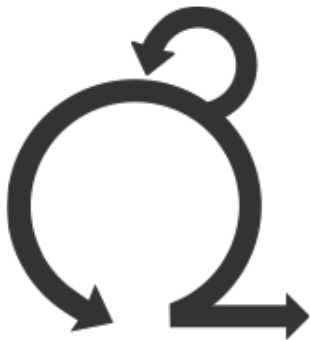
- ▶ Agile process for small-to-medium-sized teams
- ▶ Decisions made by consensus
- ▶ No specializations among the programmers
- ▶ Developers have only the programmer role

Model of AIP

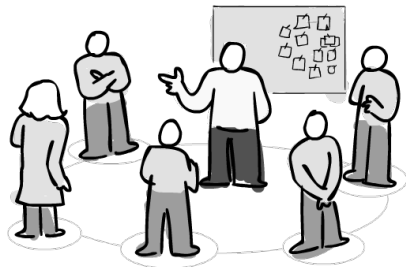


Iterations

- ▶ Iteration meeting
- ▶ Assessing current state of the product
 - ▶ all stakeholders participate
 - ▶ technical and business point of view
- ▶ Planning the next iteration
 - ▶ iteration backlog
 - ▶ extracted from product backlog



Daily meeting



- ▶ Daily problems and challenges
- ▶ A consensus about the progress
- ▶ Daily assignments of change request
- ▶ Clarify the ambiguities
- ▶ Needs for code refactoring
- ▶ Early warning when anything goes wrong

Agile Manifesto

- ▶ Developed in 2001 (17 original authors)
- ▶ Signed by numerous people since

Individuals and interactions over processes and tools

Working software over comprehensive documentation

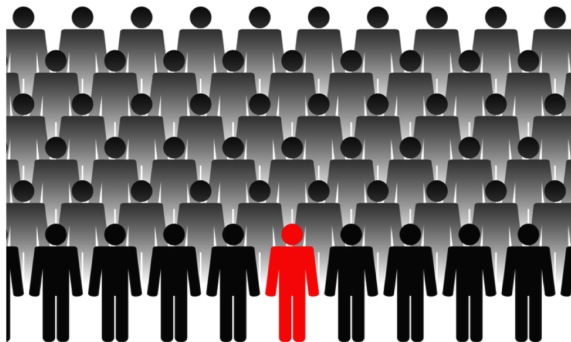
Customer collaboration over contract negotiation

Responding to change over following a plan

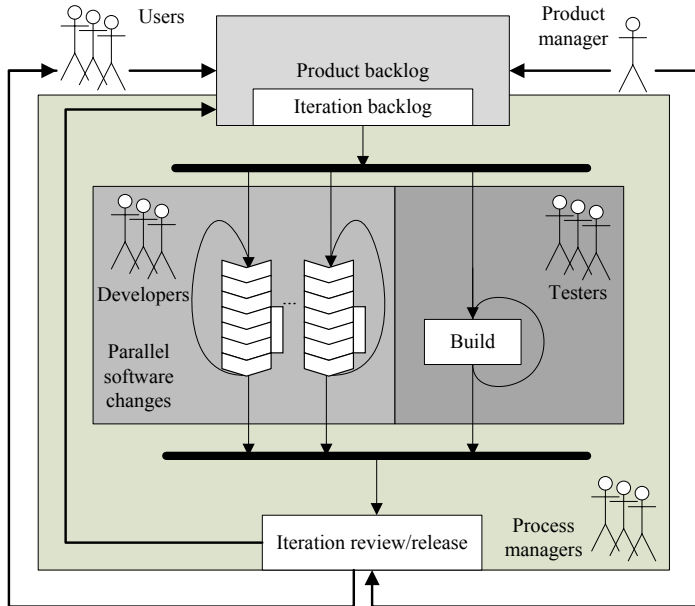
Directed Iterative Processes (DIP)

DIP

- ▶ Process runs under direction of managers
- ▶ Several different specialized roles for the programmers
- ▶ The process scales to large teams and large systems



Model of DIP



The Roles

- ▶ Developers produce code
- ▶ Testers verify new baseline
- ▶ There can be additional specialized roles
- ▶ Specialization increases effectiveness



Architect



- ▶ Guarantees that developers preserve software architecture constraints
- ▶ Approves or disapproves commits

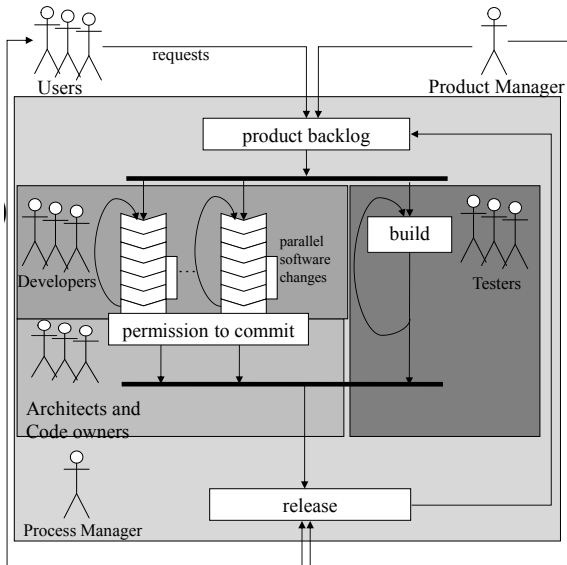
Management

- ▶ Product managers make strategic decisions
- ▶ Process managers assign tasks and control the process



Centralized Iterative Processes (CIP)

Model of CIP



Code ownership

- ▶ Programmers specialize in certain parts of the code
- ▶ Coordination can become a problem



Open Source Development



- ▶ Safeguarded
- ▶ Code ownership
- ▶ Wide community of developers, variable skills